Reliable adaptation in Hutch++

When is a change worth trusting?

$$\mathbf{v} \longmapsto A\mathbf{v}, \qquad \text{estimate } \operatorname{tr}(A) \text{ with } m \text{ queries.}$$

One query can be an expensive Hessian–vector product.

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Hutchinson: average random trace samples

Random sign probe $g_j \in \{-1, +1\}^d$ → One matrix query Ag_j → One scalar sample $g_j^\top Ag_j$

$$\hat{t}_s = \frac{1}{s} \sum_{j=1}^{s} g_j^\top A g_j$$

$$\mathbb{E}[\hat{t}_s] = \operatorname{tr}(A).$$

Hutch++ uses the same averaging on a smaller residual.

Hutch++: sketch, then sample the residual

Meyer et al. (2021); adaptive variants: Persson et al. (2022).

Build a basis Q
q sketch queries
→
Evaluate the basis
r basis queries
→
Sample the residual
ℓ final probes

$$q + r + \ell = m, \qquad \underbrace{q = r = \ell = m/3}_{\text{standard full-rank split}}.$$

Our focus: reliable decisions within a fixed total budget.

Can we trust an allocation change?

$$\Delta = \underbrace{\mathcal{R}_a}_{\text{MSE of proposed allocation}} - \underbrace{\mathcal{R}_0}_{\text{MSE of baseline}} \qquad \text{Is } \Delta < 0?$$

- Freeze both choices before gathering checking data.
- Count construction, checking, and final sampling.

A lower estimated risk still needs an uncertainty argument.

Comparing residual risks directly

$\widehat{\sigma}_x^2$: residual sample variance; use the same fresh probes for both choices.

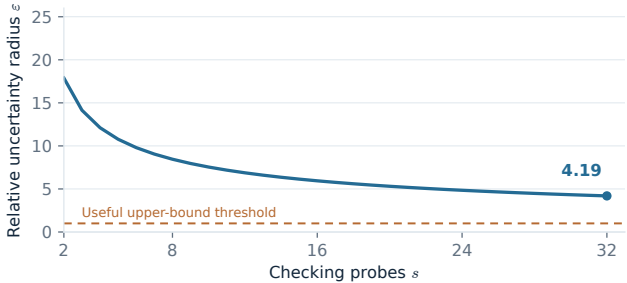
$$\widehat{\Delta} = \frac{\widehat{\sigma}_a^{\,2}}{\ell_a} - \frac{\widehat{\sigma}_0^{\,2}}{\ell_0}.$$

The comparison is unbiased: $\mathbb{E}[\widehat{\Delta}] = \Delta$.

Positive covariance can reduce comparison variance.

A valid bound can still be too wide

A conservative 95% uncertainty bound for two fixed choices, given by Chebyshev.

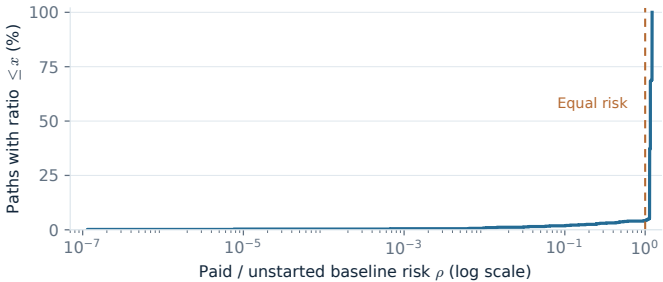


For $2 \leq s \leq 32$, the radius exceeds 1: the bound is inconclusive.

This limits the specified bound, not every certification method.

Checking can spend away the benefit

Sample-variance heuristic: $m = 160$, $s = 16$; all query costs charged.



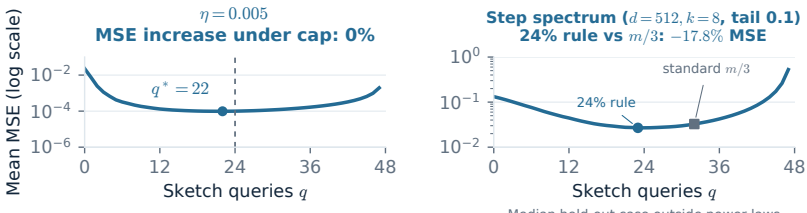
575 of 600 paths had higher average policy risk after checking.

Can structure guide the choice before we spend on checking?

A smaller sketch than $m/3$ helps across many structures

- Fixed $q \approx m/4$ vs standard $q = m/3$, same total budget
- 186 held-out matrices, 16 groups: $\approx 20\%$ lower mean normalized MSE
- Near-optimal for power laws at tested $\eta = 0.005, 0.02$; fails at very small floors

$m = 96$. Left: power law ($a = 1.8$, $d = 512$), cap $q \leq 24$. Right: held-out $q = 23$ vs. $q = 32$.



$m/3$ oversketches in many cases; how much smaller is safe depends on matrix structure.

Ratio chosen on training groups; $q = \lfloor 0.24m + 0.5 \rfloor$. A predefined 15% rule did slightly better (22%). Means over cases; individual matrices can be hurt.

The missing link: a cheap, reliable choice

Recent work: locate promising allocations.

Earlier work: study confidence and decision costs.

Neither gives a cheap online selection rule yet.

What we learned, and what comes next

- Reliable decisions need more than unbiased risk estimates.
- Small sketches can work; the conditions depend on matrix structure.
- Next: find cheap structural clues without spending away the benefit.

Limits: selected synthetic matrices and budgets; no universal cap or completed online selector.